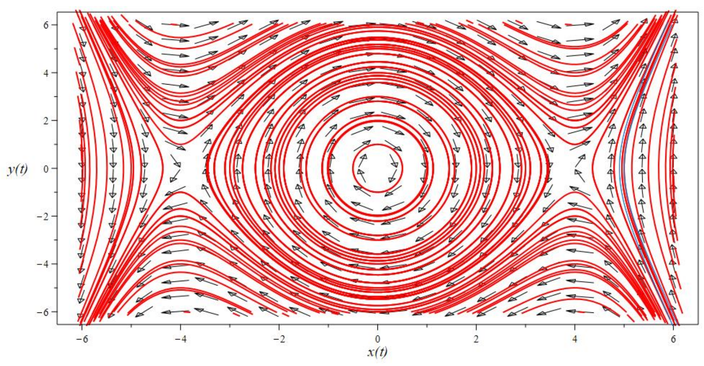
Undergraduate Research

at the School of Mathematical and Statistical Sciences, University of Galway

This poster presents a small selection of the images created by undergraduate students in the School of Mathematical and Statistical Sciences, as part of their final year projects. They show the diversity of topics our students are researching, covering numerous areas of pure and applied mathematics, statistics, bioinformatics, and computing.

Thomas O'Connor: *The Duffing Equation*



Dara Kelly and Cronan Ward: *Modelling Extremes in Finance and Investigating Asymptotic Dependence*

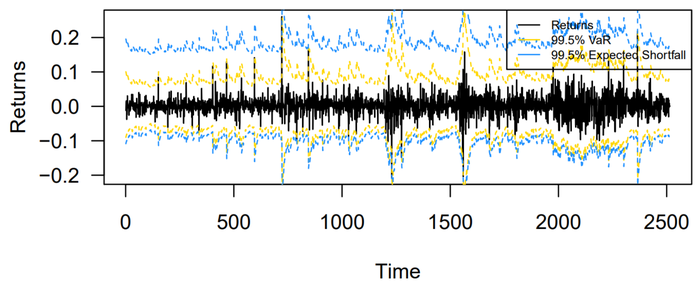
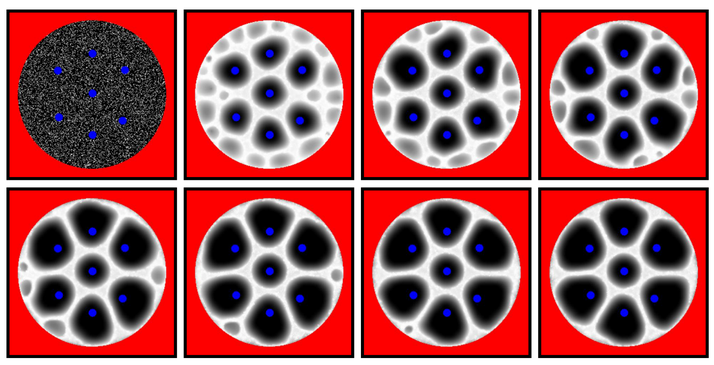
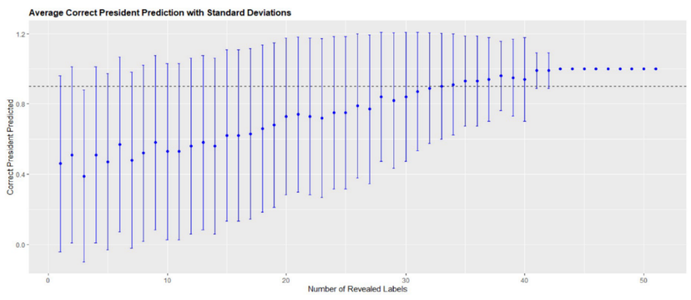


Figure 25: NVDA returns with VaR 99.5% and Expected Shortfall 99.5% Bounds (1st Jan 2014−2024)

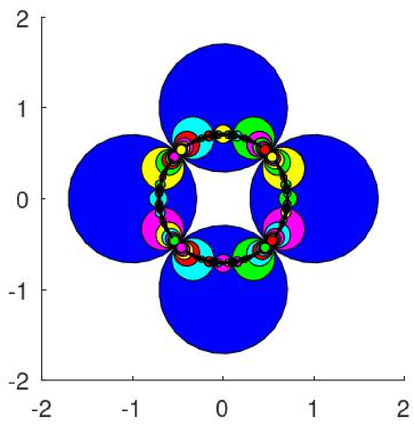
Sean Janson and Dmytro Lyubka: *Unveiling Emergent Patterns and Euclidean Approximations using Physarum Polycephalum*



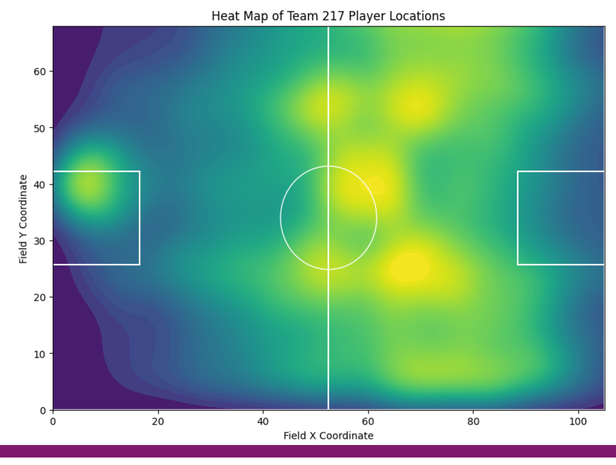
Lise Wall and Emma Meaney: *Predicting the US president election using network approaches*



Andrew Bell and Shane Cotter: *Fractals and Mobius Maps*



Ailbhe Keane, Karen Gillooly and Paulina Krungleviciute: *Analysis of Expected Goals (xG) of Team 217 during La Liga 2019/2020 season*



Katelyn McMahon and Matthew Talbot: *The Impact of County Demographics and Major Events on Votes for the Democratic Party*

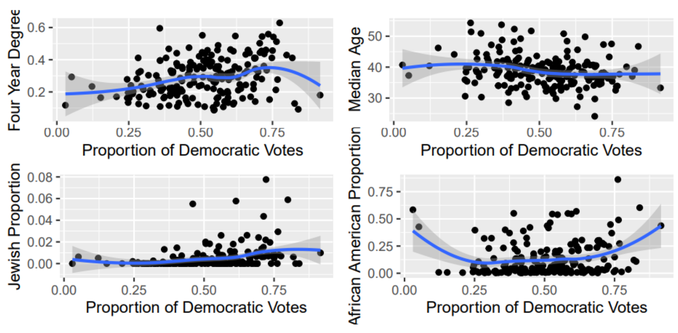
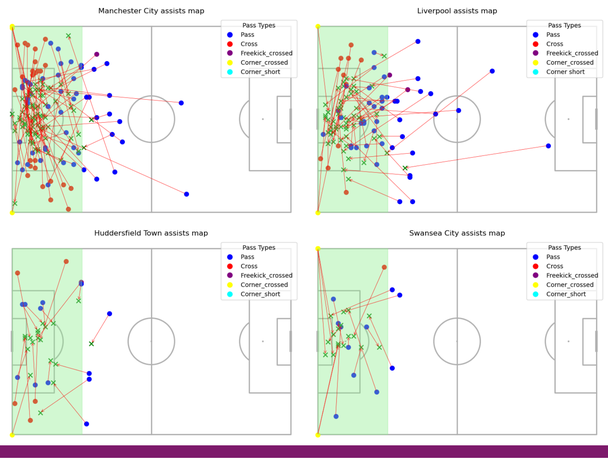


Figure 3: Relationship between the proportion of Democratic votes in a senatorial election and some predictor variables.

Cian Hughes and Conal O'Mahony: *Investigating the evolution of tactics in soccer through data analysis*



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